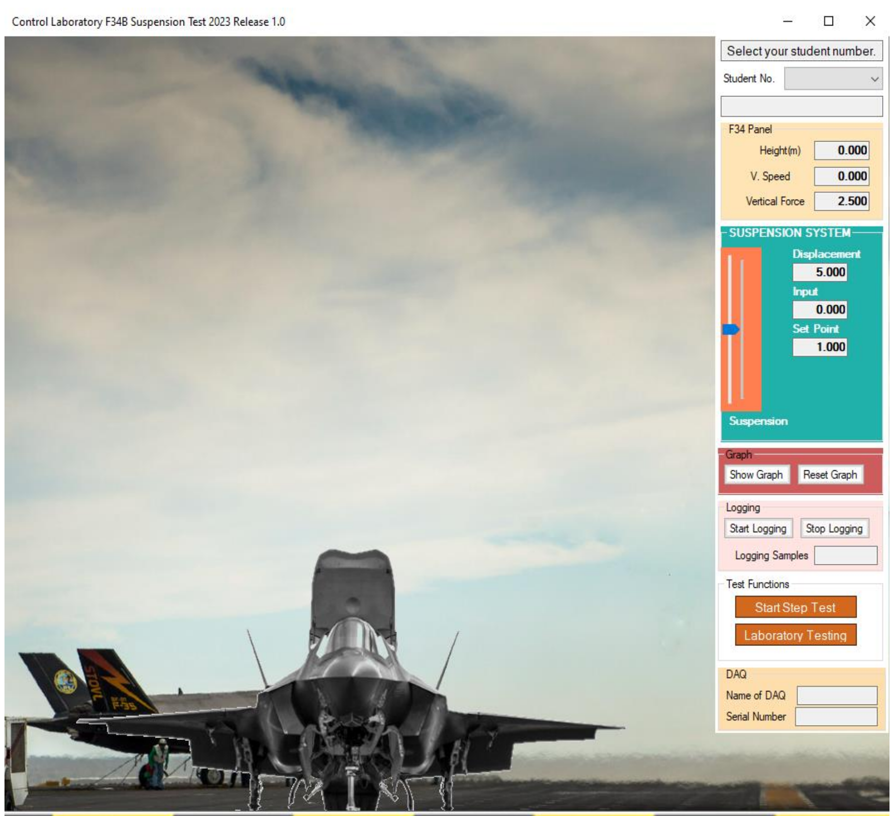
UCT CONTROL LABORATORY

User Manual

Control Laboratory Suspension Simulator

Release 1.0

10-Aug-23



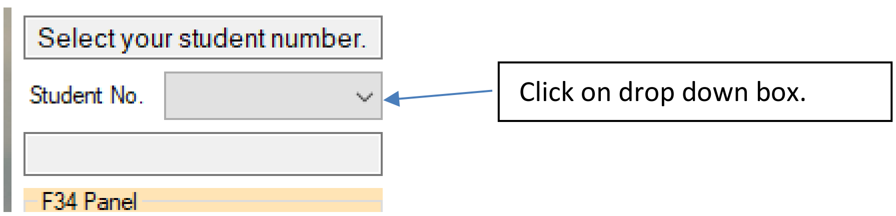
Contents

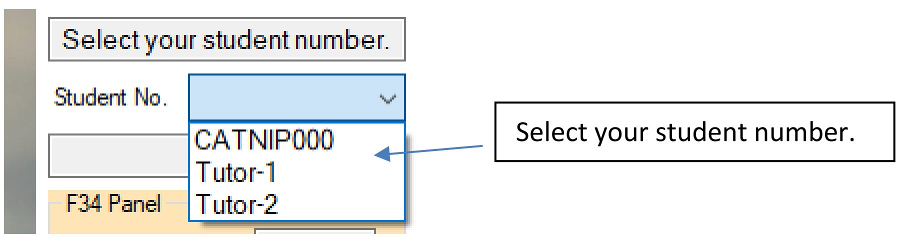
1. Description

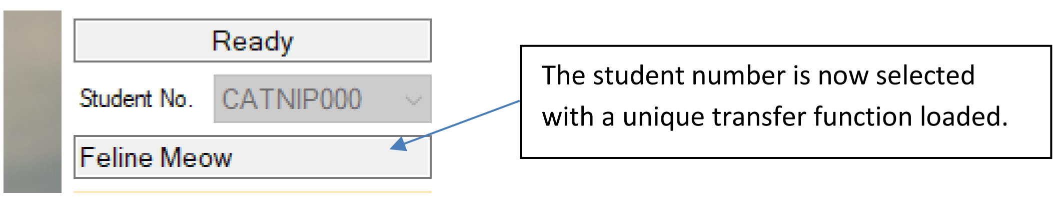
The Control Laboratory suspension simulator simulates the shock absorption of a landing vertical take off jet fighter. The simulator uses an image of a F-34B Jet Fighter vertical takeoff and landing to demonstrate shock absorption. The purpose of the simulator is to generate a function that results in the displacement of the suspension when a constant mass is applied.

2. Student Sign On

The generated functions are unique to the student numbers. Hence students need to sign on before doing any testing. Once you have started the software by clicking on the icon do the following.

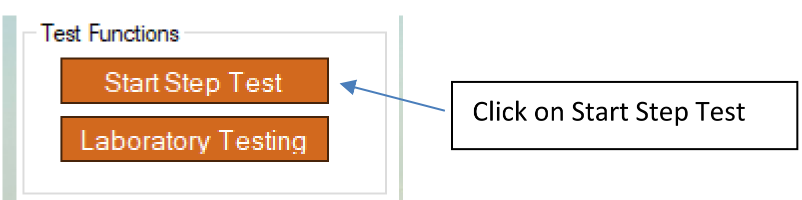


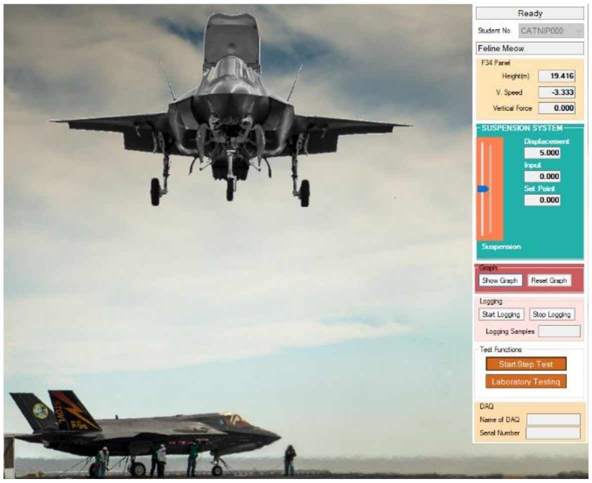




3. Start Step Test

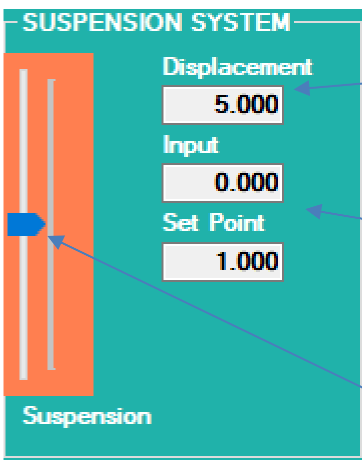
To start a "realistic" step test click the Start Step Test as shown below. The image moves up and then makes a landing generating the shock absorption response.





F34B Image moves up and then makes a landing.

The response is shown on the slider in the panel SUSPENSION SYSTEM shown below.



Shows the value of the displacement.

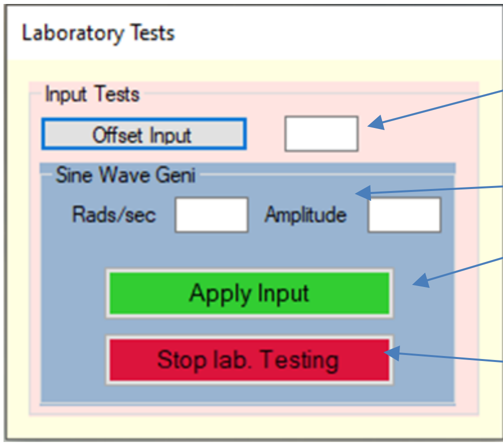
Displays the Input and the Set Point during testing.

Indicates the displacement of the mass on the suspension system.

4. Laboratory Testing

Laboratory testing is used to subject the suspension system to known inputs so that the output responses may be recorded. This data is then used to derive a mathematical model of the system.

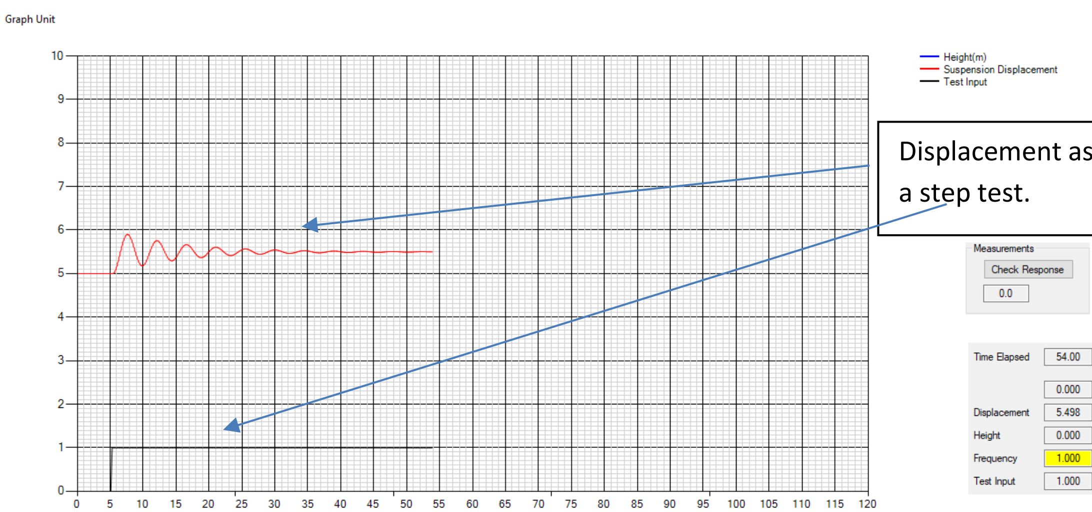
Click on Laboratory testing to start with controlled input testing.



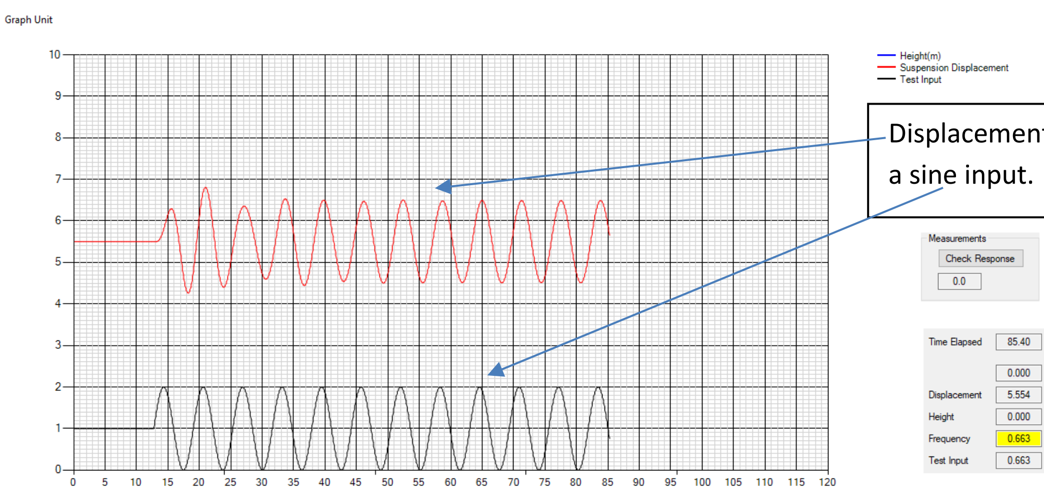
Offset Input is used to a invoke a step input of a certain size.

A sine wave can be generated by entering the frequency in radians with a certain amplitude.

Stops the laboratory testing.

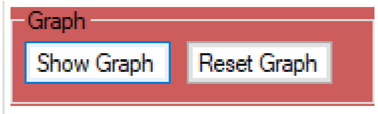


Displacement as a result of a step test.

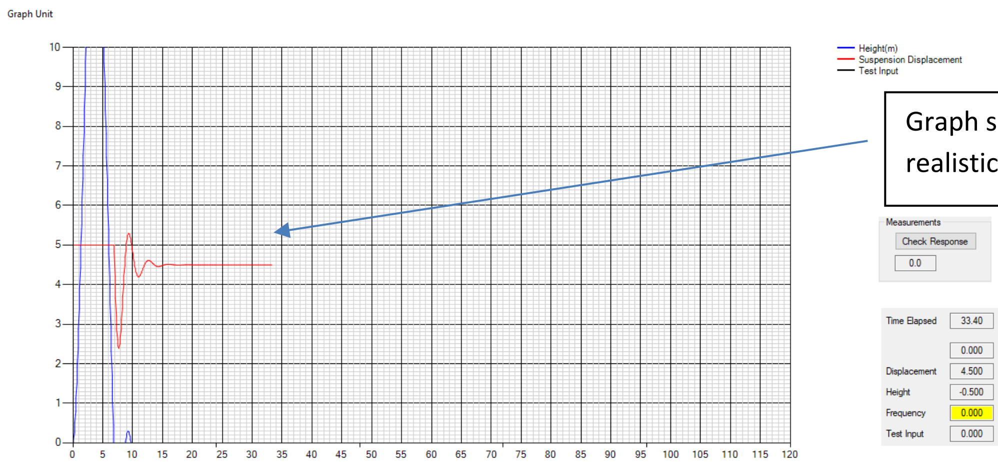


Displacement as a result of a sine input.

5. Show Graph and Reset Graph

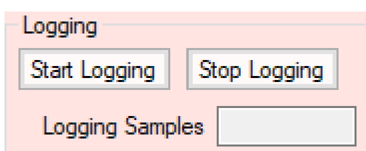


Click on Show Graph to view the testing graphically. Reset Graph is to start the graph plotting from zero.

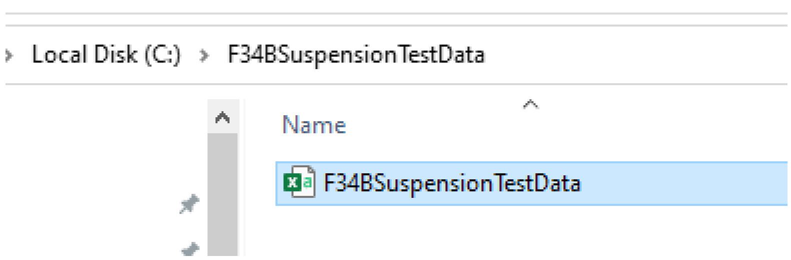


Graph showing the response of the realistic step test.

6. Logging Data



Start logging records the response. Stop logging stops the logging.



The data is saved in the folder name F34BSuspensionTestData and a file name of F34BSuspensionTestData. The file format is in csv.

	A	B	C	D	E
1	Time(s)	Input	Output_Displacement		
71	137.4	1	5.179		
72	137.5	1	5.281		
73	137.6	1	5.395		
74	137.7	1	5.514		
75	137.8	1	5.635		
76	137.9	1	5.752		
77	138	1	5.861		
78	138.1	1	5.96		
79	138.2	1	6.045		
80	138.3	1	6.116		
81	138.4	1	6.171		
82	138.5	1	6.21		
83	138.6	1	6.233		
84	138.7	1	6.241		
85	138.8	1	6.236		
86	138.9	1	6.219		
87	139	1	6.192		
88	139.1	1	6.157		
89	139.2	1	6.116		
90	139.3	1	6.072		
91	139.4	1	6.027		
92	139.5	1	5.982		
93	139.6	1	5.939		
94	139.7	1	5.899		
95	139.8	1	5.864		

Data is logged at a sample time of 100ms. The Input, Output Displacement are logged.